



Smart City Mobility and Air Quality Analysis

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Why Smart City Mobility & Air Quality Matter ?

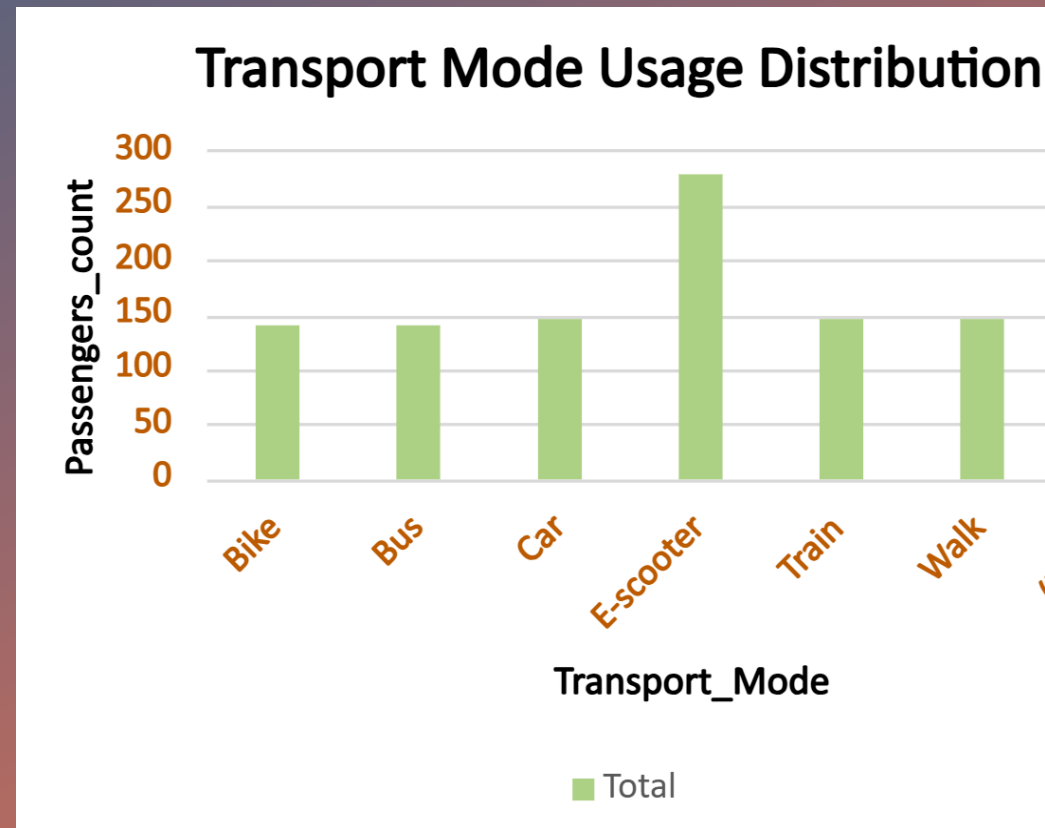


Objectives:

- Analyse mobility patterns
- Evaluate transport efficiency
- Assess the impact of weather on journeys
- Measure environmental impact through CO₂ emissions and Air Quality Index
- Identify factors affecting passenger satisfaction

Mobility Patterns: Transport mode usage distribution

How do people travel
around the city ?



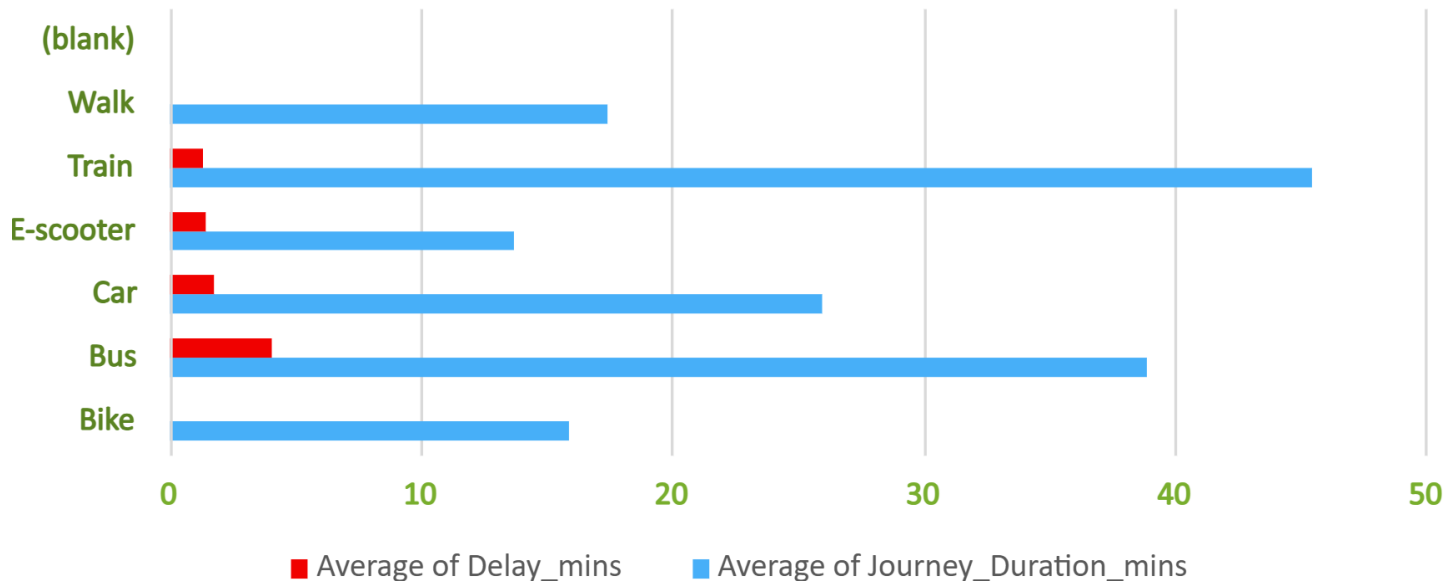
E-scooter is the most used transport mode, while bike is the least used transport mode.

Similar usage level for trains, Bus, Car, Walk, and Bike

Mobility Performance:

Which transports mode are the most efficient?

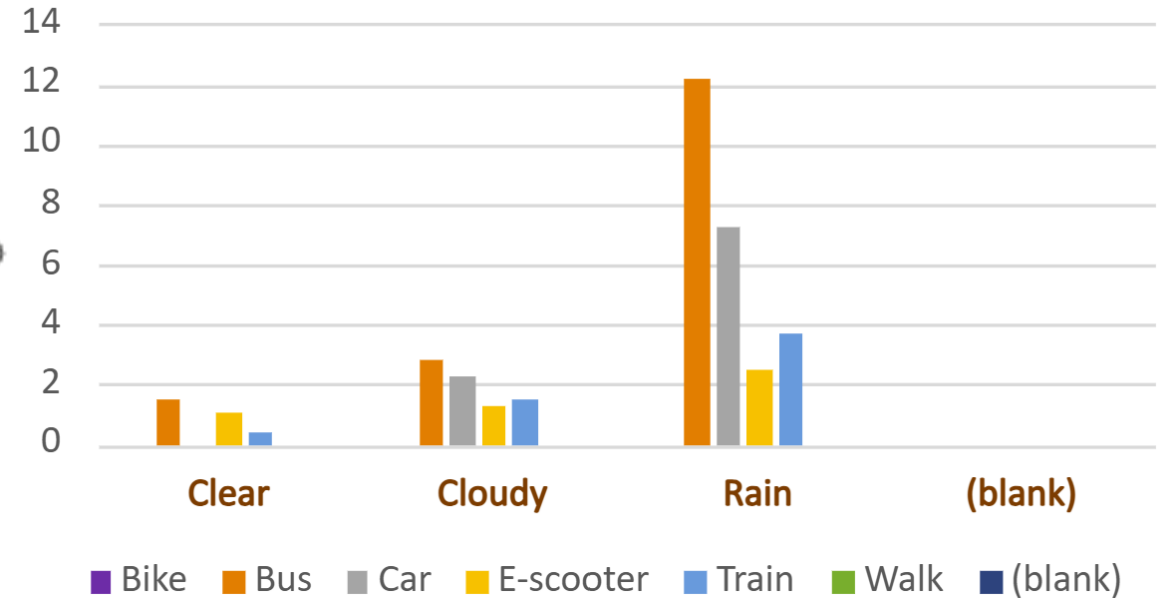
Transport Mode Performance: Duration vs Delay



- **E-scooters, Bike and Walking are the most efficient and reliable modes: Ideal for short trips.**
- **Bus and Cars are the most affected by delays: Traffic congestion**
- **Trains are used for long journeys, and reliable due to short delays.**

How does weather impact on Transport Efficiency?

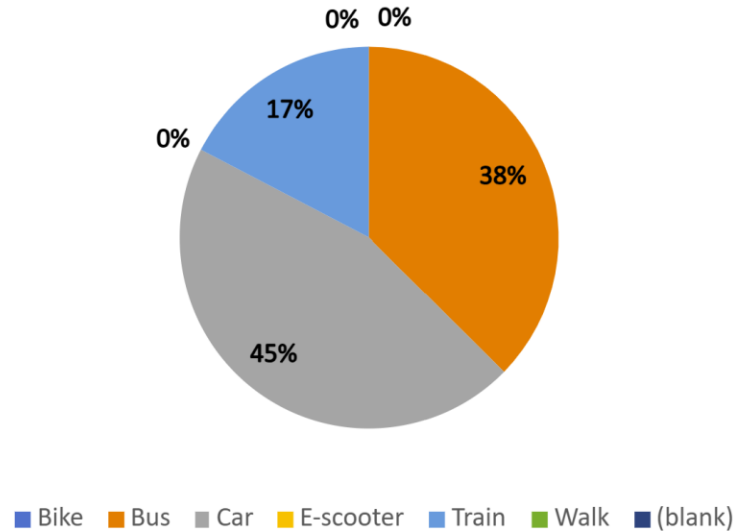
Weather vs Transport Performance



- Rain significantly increases delays time across all transport modes, especially on Buses and cars.
- E-scooters and trains are slightly affected by weather.
- Walking and cycling are not affected by weather.

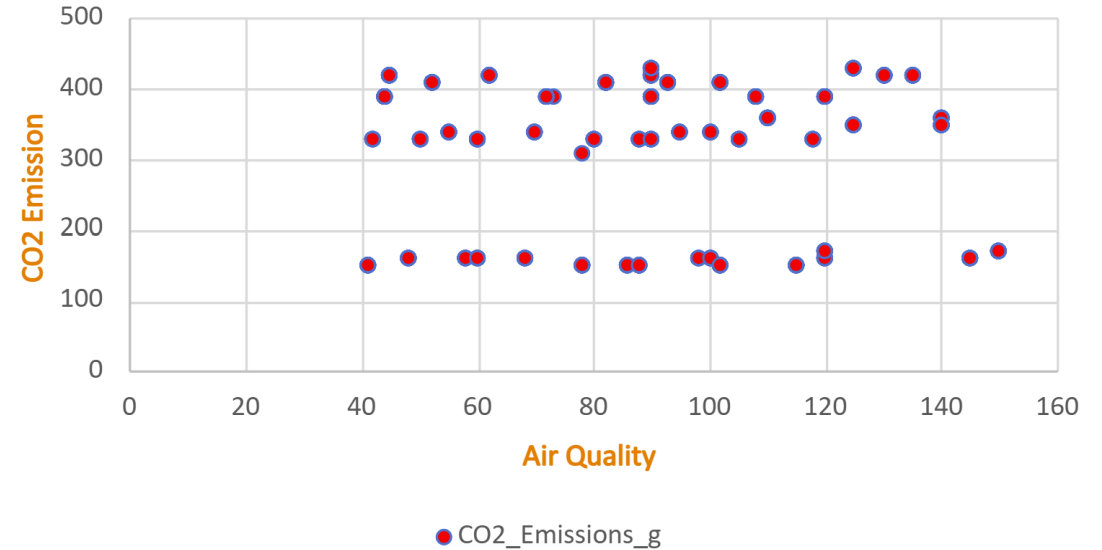
Air Quality and Sustainability

Average CO₂ Emission vs Transport Mode



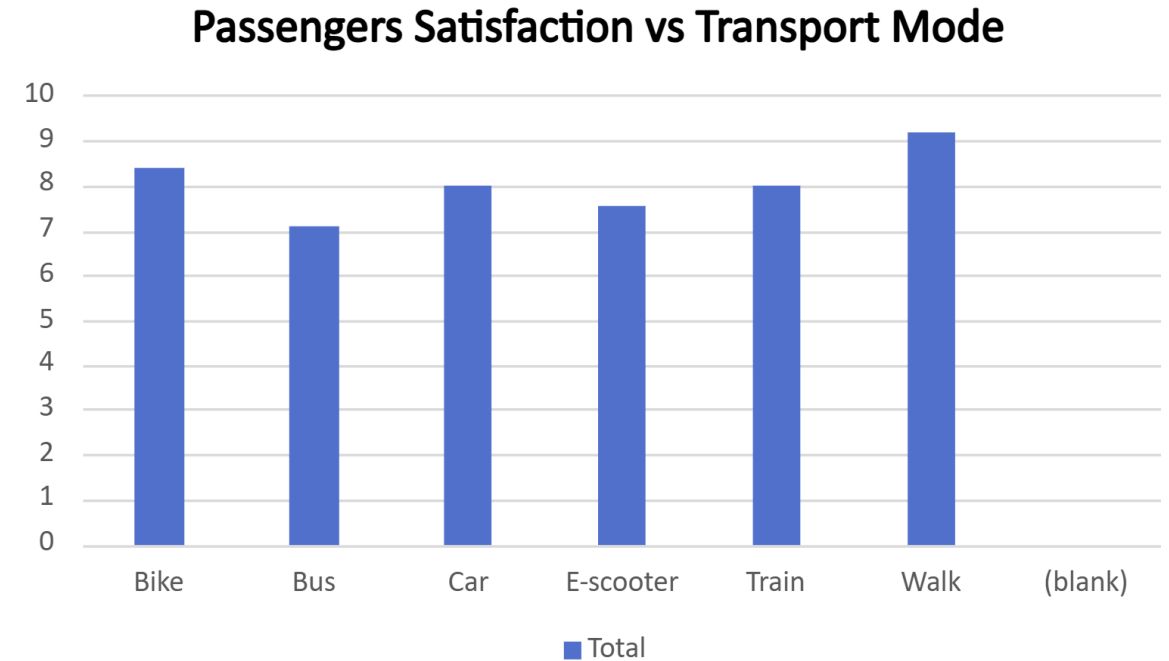
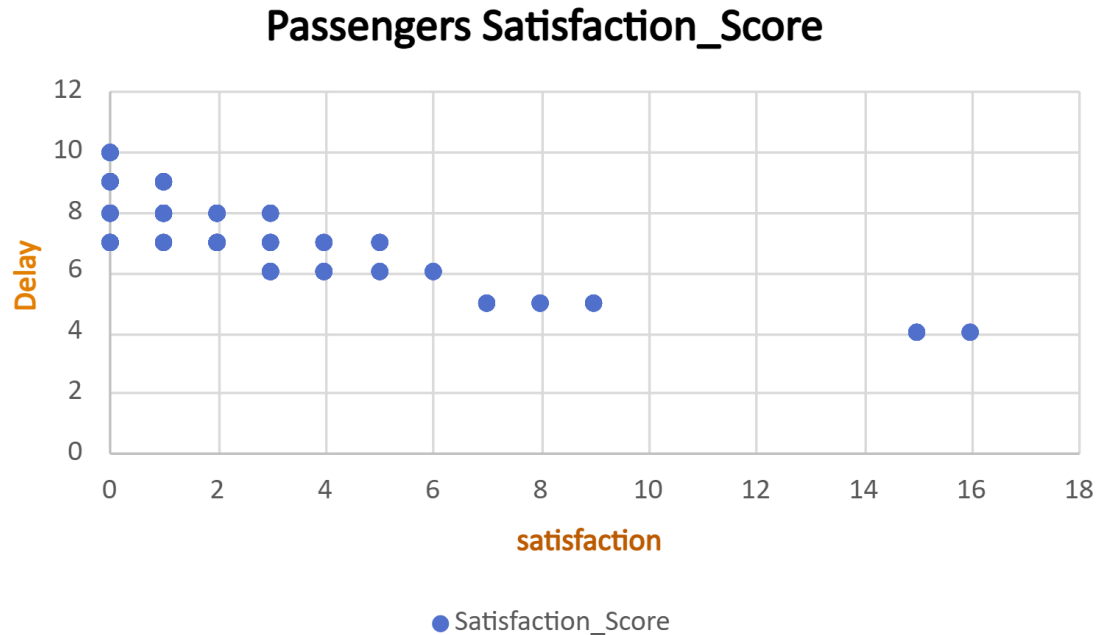
- Cars produce the highest CO₂ emissions
- Buses and trains generate significantly lower emissions
- Walk, Bike, and E-scooters produce zero emissions

CO₂_Emissions_g vs Air Quality



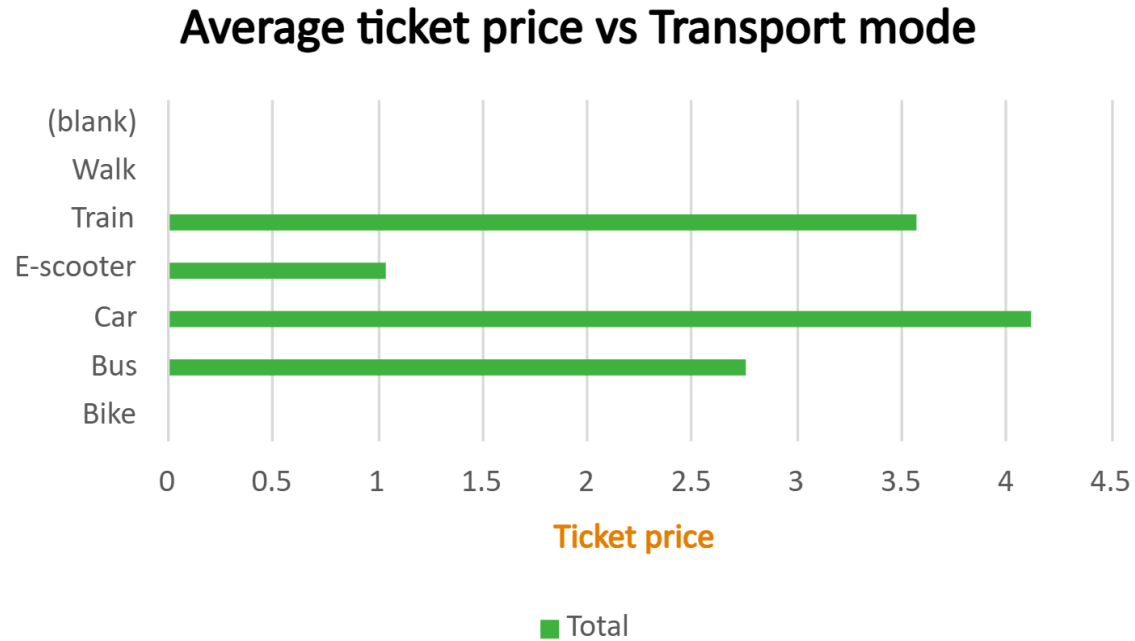
- Higher emissions are associated with poorer air quality.
- Cars, Buses and Trains contribute to the poor air quality.

Passengers Satisfaction



- Walking has the highest satisfaction score
- Bus travel has the lowest satisfaction
- Sustainable and flexible transport modes (bike, train, car, e-scooter) show relatively high satisfaction

Ticket Price Analysis



- **Walking and cycling are completely free.**
- **E-scooters are the cheapest paid option**
- **Cars and trains are the most expensive modes**

Conclusion

- **The choice of Transport mode can be influenced by delay time, weather, and passenger satisfaction scores.**
- **Cars generate the highest CO₂ emissions.**
- **Walking is the most efficient and sustainable transport mode**
- **Bus has the highest delay time.**

Recommendations:

- **Improve the bus frequency.**
- **Encourage electric buses, trains and cars.**
- **Encourage the use of Bike and E-scooter through bike to work scheme.**

**THANK YOU
FOR YOUR
ATTENTION**

